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DISPLAY DEVICE

Abstract

A display device includes a display section and a control section, and displays an operation to be executed by a user in a component mounting line on a process-by-process basis on the display section. The control section is configured to, when a first image illustrating a content of a first process to be currently executed by the user is being displayed on the display section and when there exists a second process to be executed by the user after the first process, display a second image for guiding the second process on the display section along with the first image.

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Background/Summary

TECHNICAL FIELD

[0001] The technique disclosed in the present description relates to a display device that displays operation contents to be executed by a user in a component mounting line.

BACKGROUND ART

[0002] JP-A-2016-225387 discloses a system for supporting component replenishment of a component feeder of a component mounter in a component mounting line. In this system, the component shortage time of each component feeder is predicted, and the shortest time among the times until the component shortage time of each component feeder reaches a predetermined warning time is displayed on a display section.

BRIEF SUMMARY

Technical Problem

[0003] In the technique of JP-A-2016-225387, a list of component feeders whose component shortage time arrives within the warning time is displayed, but is not taken into consideration in the operation currently executed by a user. The present description proposes a technique capable of more clearly knowing contents of processes to be executed by a user.

Solution to Problem

[0004] A display device disclosed in the present description includes a display section and a control section, and displays an operation to be executed by a user in a component mounting line on a process-by-process basis on the display section. The control section is configured to, when a first image illustrating a content of a first process to be currently executed by the user is being displayed on the display section and when there exists a second process to be executed by the user after the first process, display a second image for guiding the second process on the display section along with the first image.

[0005] In the above display device, the first image illustrating the content of the first process to be currently executed by the user (that is, a process currently being executed) along with the second image for guiding the second process to be executed by the user after the first process are displayed on the display section. Therefore, the user can perform the operation while checking the content of the process currently being executed, and can easily know what process the user needs to execute next.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0006] FIG. 1 is a schematic diagram illustrating a configuration of a component mounting system.

[0007] FIG. 2 is a schematic diagram illustrating a configuration of a component mounter.

[0008] FIG. 3 is a cross-sectional view taken along line III-III of FIG. 2.

[0009] FIG. 4 is a block diagram illustrating configurations of the component mounter and a management device.

[0010] FIG. 5 is a diagram illustrating an example of a display format of contents to be executed by a user.

[0011] FIG. 6 is a diagram illustrating another example of the display format of the contents to be executed by the user.

[0012] FIG. 7 is a diagram illustrating still another example of the display format of the contents to be executed by the user.

[0013] FIG. 8 is a diagram illustrating yet still another example of the display format of the contents to be executed by the user.

[0014] FIG. 9 is a diagram illustrating an example of a display format of a detailed image.

[0015] FIG. 10 is a diagram illustrating an example of a display format of an image illustrating contents of processes for resolving an error.

DESCRIPTION OF EMBODIMENTS

[0016] Main features of an example, which will be described below, will be listed. Technical elements, which will be described below, are each independent technical elements and are designed to exhibit technical usefulness alone or by being combined together in various ways, and the combinations are not limited to those described in claims.

[0017] In one embodiment of the present technique, the second image may illustrate only a part of a content of the second process.

[0018] In such a configuration, only the part of the content of the second process is displayed on the display section. Therefore, while the user can know the existence of the second process to be executed after the first process, the user can easily concentrate on the first process to be currently executed without being distracted by the existence of the second process.

[0019] In one embodiment of the present technique, the part of the content of the second process may be one or multiple elements selected from a process name, an operation name, an operation location, an operation start time, and an operation content of the second process.

[0020] In such a configuration, the user can more specifically know the content of the second process.

[0021] In one embodiment of the present technique, the control section may be configured to, as an operation start time of the second process approaches, change and display a content of the second image in a format in which the user more easily recognizes the content of the second image.

[0022] In such a configuration, the user can gradually and easily recognize a guidance of the second process included in the second image. In other words, as the operation start time of the second process approaches, the user can know many contents related to the second process, and can smoothly transition to the second process.

[0023] In one embodiment of the present technique, the control section may be configured to, as the operation start time of the second process approaches, increase the content of the second process to be displayed as the second image.

[0024] In such a configuration, since the amount of information (for example, display items) related to the second process increases as the operation start time of the second process approaches, the user can smoothly transition to the second process.

[0025] In one embodiment of the present technique, the control section may be configured to, when a third process which is different from the second process and which is to be executed by the user exists after the first process and either simultaneously with or after the second process, display a third image for guiding the third process on the display section along with the first image and the second image.

[0026] In such a configuration, when there exists the third process to be executed simultaneously with the second process or after the second process, the user can perform the operation while checking the content of the process currently being executed and also know the existence of the third process.

[0027] In one embodiment of the present technique, the control section may be configured to, when the operation start time of the second process and an operation start time of the third process are the same, assign priority, which is to be executed preferentially, to the second process and the third process, and display an image for guiding a process assigned a first priority in a format in which the user more easily recognizes the image for guiding the process assigned the first priority than an image for guiding a process assigned a second priority.

[0028] In such a configuration, the process to be executed first among the second process and the third process is displayed in a format that is easily recognizable, so that the user can easily know what is to be executed after the process currently being executed.

[0029] In one embodiment of the present technique, the second image may include a first selection image for the user to select to execute the second process before the first process, and the control section may be configured to, when the first selection image is selected by the user, display the

second image in a format in which the user more easily recognizes the second image than the first image.

[0030] In such a configuration, for example, when the user determines that the second process needs to be executed before the first process, the user can know the content of the second process in more detail by selecting the first selection image.

[0031] In one embodiment of the present technique, the control section may be configured to, when the first image and the second image are being displayed on the display section, further display a second selection image for the user to select to display a detailed image illustrating a list of operations to be executed by the user in the future on the display section, and when the second selection image is selected by the user, display the detailed image on the display section.

[0032] In such a configuration, what kind of operations the user is scheduled to perform in the future is displayed as a list. Therefore, the user can know the future operation contents at once.

[0033] In one embodiment of the present technique, the control section may be configured to, when an error occurs in the component mounting line while the first image is being displayed on the display section, display a fourth image illustrating a content of a process to be executed by the user to resolve the error on the display section in a format in which the user more easily recognizes the fourth image than the second image.

[0034] When an error occurs in the component mounting line, it is preferable to quickly resolve the error. In the above configuration, when an error occurs in any part of the component mounting line while executing the first process, the content of the process to be executed to resolve the error is displayed in a format in which the user more easily recognizes the content of the process to be executed to resolve the error than the content of the second process to be executed after the first process. Accordingly, the user can know that an error has occurred, and can quickly address the error.

EXAMPLE

[0035] Hereinafter, component mounting system **1** of an example will be described. As illustrated in FIG. **1**, component mounting system **1** includes two component mounting lines **2** and management device **8**. Each component mounting line **2** has a function of manufacturing a mounting board by mounting electronic components **4** on board C, and mainly includes solder printer **5** and multiple component mounters **6**. Solder printer **5** is a device that prints cream solder on board C. Component mounter **6** is a device that mounts electronic components **4** on board C on which the cream solder is printed. Although not illustrated in FIG. **1**, component mounting line **2** is further provided with a print inspection machine which is disposed downstream of solder printer **5** and inspects the state of the cream solder printed on board C, a visual inspection machine which is disposed downstream of component mounter **6** and inspects the state of board C on which electronic components **4** is mounted, a reflow oven which is disposed downstream of the visual inspection machine and heats board C to reflow solder electronic components **4**, and the like. Management device **8** controls operations of various devices incorporated in each component mounting line **2**, and manages mounting of electronic components **4** on board C in each component mounting line **2**.

[0036] Next, the configuration of component mounter **6** will be described. As illustrated in FIGS. **2** to **4**, component mounter **6** includes component mounting section **22**, interface device **24**, and control device **30**. Component mounting section **22** includes multiple component feeders **12**, feeder holding section **14**, mounting head **16**, head moving device **18**, and board conveyor **20**.

[0037] Each component feeder **12** accommodates multiple electronic components **4**. Component feeder **12** is detachably attached to feeder holding section **14** and supplies electronic components **4** to mounting head **16**. A specific configuration of component feeder **12** is not particularly limited. Each component feeder **12**, for example, may be any of a tape-type feeder for accommodating multiple electronic components **4** on a wound tape, a tray-type feeder for accommodating multiple electronic components **4** on a tray, or a bulk-type feeder for accommodating multiple electronic

components **4** randomly in a container.

[0038] Feeder holding section **14** includes multiple slots, and component feeder **12** can be detachably installed in each slot. Feeder holding section **14** may be fixed to component mounter **6** or may be detachable from component mounter **6**. Mounting head **16** detachably holds one or multiple nozzles **7**, uses nozzle **7** to pick up electronic components **4** supplied by component feeder **12**, and mounts electronic components **4** on board C. At this time, head moving device **18** moves mounting head **16** with respect to component feeder **12** and board C. Accordingly, electronic components **4** are picked up from specific component feeder **12** of multiple component feeders **12**, and electronic components **4** are mounted on a position determined on board C in advance. Board conveyor **20** conveys in, supports, and conveys out board C.

[0039] Interface device **24** is a device that provides the user with various information on component mounter **6** and is a device that receives instructions and information from the user. In the present example, interface device **24** is a touch panel, but may include, for example, a display and a keyboard or a mouse.

[0040] Control device **30** is communicably configured with management device **8**. Control device **30** controls the operation of each element of component mounter **6** in response to instructions from management device **8**. As illustrated in FIG. **4**, control device **30** is configured with a computer including memory **32** and CPU **34**. Control device **30** refers to mounting data **36** (received from management device **8**) stored in memory **32** to control the operation of component mounting section **22** and the like, and to mount electronic components **4** on board C. In addition, CPU **34** includes detection section **40** that detects state data **38** indicating the state of the mounting process in the process of executing the mounting process on board C. In memory **32**, state data **38** detected by detection section **40** is stored in real time. State data **38** includes, for example, mounting line data for identifying each component mounting line **2**, mounter data for identifying each component mounter **6**, slot data indicating a position of a slot in which each component feeder **12** is mounted, component type data indicating a component type accommodated in each component feeder **12**, and component shortage time data indicating a time until each component feeder **12** runs out of components. Control device **30** transmits state data **38** to management device **8** at predetermined intervals.

[0041] Next, the configuration of management device **8** will be described. As illustrated in FIG. **4**, management device **8** includes interface device **52** and arithmetic device **60**.

[0042] Interface device **52** is a device that provides the user with various information on component mounting line **2** and is a device that receives instructions and information from the user. In particular, as will be described later, interface device **52** of the present example functions as a device that notifies the user of contents of operations to be executed by the user in the mounting process in component mounting line **2**. In the present example, interface device **52** is a touch panel, but may include, for example, a display and a keyboard or a mouse.

[0043] Arithmetic device **60** is configured with a computer including memory **62** and CPU **64**. Production program **66** is stored in memory **62**. The operation of entire component mounting line **2** is described in production program **66**, and arithmetic device **60** instructs the various devices incorporated in component mounting line **2** to perform the operation to be executed by each device according to production program **66**. Memory **62** stores multiple state data **38** transmitted from each control device **30** of each component mounter **6**. CPU **64** includes execution order determination section **68** that determines the order of operations to be executed by the user. Based on multiple state data **38** received from each component mounter **6**, execution order determination section **68** determines the order of operations to be executed by the user in the production of board C in component mounting system **1**.

[0044] Next, in the production of board C in component mounting system **1**, the process in which arithmetic device **60** displays contents of operations to be executed by the user on interface device **52** will be described. When board C is produced in component mounting line **2**, there exist multiple

operations to be executed by the user for each component mounter **6**. Examples of the contents of the operations include a component replenishment operation (that is, an exchange operation of component feeder **12**) caused by a decrease in the number of components accommodated in component feeder **12** and an exchange operation of nozzle **7** caused by a change in the type of electronic components **4** to be mounted. In the present example, arithmetic device **60** displays these operations to be executed by the user on a process-by-process basis on interface device **52**. In the present description, both terms are used so that “operation” has a broader concept than “process”. That is, one or multiple “processes” exist for one “operation”.

[0045] As processes in the exchange operation of component feeder **12**, there are processes such as carrying component feeder **12** accommodating the electronic components of the designated type to component mounter **6** targeted for exchange, replacing component feeder **12**, and returning detached component feeder **12** to a predetermined position. As processes in the exchange operation of nozzle **7**, there are processes such as carrying a nozzle holder in which nozzle **7** of the designated type is accommodated to component mounter **6** targeted for exchange, replacing the nozzle holder, and returning the detached nozzle holder to a predetermined position.

[0046] First, arithmetic device **60** determines the order of operations to be executed by the user based on state data **38** received from each component mounter **6** and production program **66**. Specifically, arithmetic device **60** calculates a time at which each component feeder **12** is to be exchanged based on the component shortage time data, a time at which nozzle **7** is to be exchanged based on production program **66**, and the like, and rearranges the target operation in ascending order of arrival time. Hereinafter, a situation in which the component shortage time of electronic component **4a** in component feeder **12** in slot **12a** of component mounter **6a** of component mounting line **2a** (refer to FIGS. **1** and **3**) arrives earliest and the exchange operation of component feeder **12** is to be currently executed will be described. In addition, it is assumed a situation that there exists an operation of exchanging the nozzle provided in component mounter **6c** with nozzle **7a** as an operation to be executed after the exchange operation of component feeder **12** in slot **12a**.

[0047] Based on state data **38**, arithmetic device **60** displays image **P1** illustrating the content of a process to be currently executed by the user on interface device **52** as illustrated in FIG. **5**. With reference to state data **38**, arithmetic device **60** displays a character string “Feeder Exchange Operation” indicating the operation name, a character string “Carry Feeder” indicating the process name, and a character string “Carry component feeder accommodating electronic component **4a** to component mounter **6a**” indicating the operation location and the operation content as image **P1**.

[0048] In addition, since there exists a process of replacing component feeder **12** in slot **12a** after the process illustrated in image **P1**, arithmetic device **60** displays image **P2** that guides the process on interface device **52** along with image **P1**. Arithmetic device **60** displays a character string “Replace Feeder” indicating only the process name as image **P2**. The process illustrated in image **P2** is a process to be executed after the process illustrated in image **P1**. Image **P2** is displayed in a smaller size than image **P1**. In the present example, image **P2** is displayed on the right side of the display area of interface device **52**.

[0049] In addition, since there exists a process of returning detached component feeder **12** after the process illustrated in image **P2**, arithmetic device **60** displays image **P3** that guides the process on interface device **52** along with images **P1** and **P2**. Arithmetic device **60** displays a character string “Return Feeder” indicating only the process name as image **P3**. Image **P3** is displayed below image **P2** in substantially the same size as image **P2**.

[0050] After the process illustrated in image **P3**, as described above, there exists the exchange operation of nozzle **7** of component mounter **6c**. Therefore, arithmetic device **60** displays image **P4** that guides the process on interface device **52** along with images **P1** to **P3**. Arithmetic device **60** displays a character string “10:30 Operation Start” indicating only the operation start time as image **P4**. Image **P4** is displayed below image **P3** in substantially the same size as image **P3**.

[0051] Image **P1** includes end button **B1** for the user to input that the process to be currently

executed has ended. When component feeder **12** is carried to component mounter **6a** according to image **P1** displayed on interface device **52**, the user touches end button **B1** included in image **P1**. Accordingly, arithmetic device **60** can detect that the process to be currently executed has ended. [0052] When detecting the input to end button **B1**, arithmetic device **60** ends the display of image **P1** and enlarges and displays image **P2** at the center of the display area of interface device **52** as illustrated in FIG. **6**. Arithmetic device **60** displays, as image **P2**, a character string “Feeder Exchange Operation” indicating the operation name, a character string “Replace Feeder” indicating the process name, and a character string “Replace feeder in slot” indicating the operation location and the operation content, in place of the format illustrated in FIG. **5**. That is, arithmetic device **60** displays, as image **P2**, an image specifically indicating the content of the process of “Replace Feeder” in place of the image (FIG. **5**) that guides the process of “Replace Feeder”. Arithmetic device **60** moves image **P3** and image **P4** upward and displays image **P3** and image **P4**. Here, arithmetic device **60** displays, as image **P4**, a character string “Nozzle Exchange Operation” indicating the operation name in addition to the operation start time illustrated in FIG. **5**. That is, arithmetic device **60** increases the display content of the process illustrated in image **P4** compared to the format illustrated in FIG. **5**.

[0053] Similarly to image **P1** illustrated in FIG. **5**, image **P2** illustrated in FIG. **6** includes end button **B2** for inputting that the process to be currently executed has ended. When component feeder **12** is replaced, the user touches end button **B2**. Accordingly, arithmetic device **60** can detect that the process to be currently executed has ended.

[0054] When detecting the input to end button **B2**, arithmetic device **60** ends the display of image **P2** and enlarges and displays image **P3** at the center of the display area of interface device **52** as illustrated in FIG. **7**. Arithmetic device **60** displays, as image **P3**, a character string “Feeder Exchange Operation” indicating the operation name, a character string “Return Feeder” indicating the process name, and a character string “Return detached feeder” indicating the operation location and the operation content, in place of the format illustrated in FIG. **6**. Arithmetic device **60** moves image **P4** upward and displays image **P4**. Here, arithmetic device **60** displays, as image **P4**, a character string “Component Mounter **6c**” indicating the operation location in addition to the operation start time and the operation name illustrated in FIG. **6**. That is, arithmetic device **60** increases the display content of the process illustrated in image **P4** compared to the format illustrated in FIG. **6**.

[0055] As described above, arithmetic device **60** displays the content of the operation to be executed by the user in order on a process-by-process basis. Specifically, arithmetic device **60** displays an image (for example, image **P2** in FIG. **5**) that guides the process to be executed after the process along with the content of the process to be currently executed by the user (for example, image **P1** in FIG. **5**). Therefore, the user can perform the operation while checking the content of the process currently being executed, and can easily know what process the user needs to execute next.

[0056] Further, in the above example, for example, in the format illustrated in FIG. **5**, images **P2** to **P4** that guide the process subsequent to the process illustrated in image **P1** are displayed, and only a part of the content of the process is displayed. Therefore, while the user can recognize that there is a process subsequent to the process currently being executed, the user can easily concentrate on the process illustrated in image **P1** without being distracted by the existence of the subsequent process.

[0057] In the above example, as illustrated in FIGS. **5** to **7**, the display content of image **P4** increases as the operation start time of the process illustrated in image **P4** approaches. Therefore, a large amount of content can be gradually obtained in relation to the process illustrated in image **P4**, and the user can smoothly transition to the process.

[0058] In the present example, component mounting system **1** includes multiple component mounting lines **2**. Therefore, in each component mounting line **2**, there may exist two processes

that start at the same time. In such a case, arithmetic device **60** assigns priority, which is to be preferentially executed, to the two processes. Then, arithmetic device **60** displays the process with a higher priority in a format in which the user more easily recognizes the process with the higher priority than the process with a lower priority (in the format illustrated in FIG. 5 and the like, upward). Here, for example, when each of two component mounting lines **2** includes a pre-process and a post-process for production of board C, arithmetic device **60** may assign a higher priority to a process in component mounting line **2** that executes the pre-process. With this configuration, the user can easily know the process to be executed first.

[0059] Further, for example, in the situation illustrated in FIG. 7, there may be a case where the user prefers to execute the process illustrated by image P4 (that is, nozzle exchange operation) before the process of returning the feeder due to a balance of operations with other users. In such a case, the user touches image P4. When an input operation to image P4 is detected, arithmetic device **60** switches the positions of image P4 and image P3 and displays the positions as illustrated in FIG. 8. Here, in place of the format illustrated in FIG. 7, arithmetic device **60** displays, as image P4, a character string “Nozzle Exchange Operation” indicating the operation name, a character string “Carry Nozzle” indicating the process name, and a character string “Carry nozzle holder of nozzle 7a to component mounter 6c” indicating the operation location and the operation content. With this configuration, convenience is improved when the user prefers to change the order of process to be executed for some reason.

[0060] While the user is executing the operation, arithmetic device **60** displays detail display button B3 on interface device **52** as illustrated in FIG. 5 and the like. Detail display button B3 is a button for displaying detailed image P5 illustrating a list of operations to be executed by the user in the future. When detecting the input to detail display button B3, arithmetic device **60** displays detailed image P5 at the center of the display area of interface device **52** as illustrated in FIG. 9. In the example illustrated in FIG. 9, detailed image P5 includes an operation name, an operation location, and an operation start time. In such a configuration, since what kind of operations the user is scheduled to perform in the future is displayed as a list, the user can know the contents of the future operations at once. The display content of detailed image P5 is not limited to the example illustrated in FIG. 9. For example, a process name may be further displayed.

[0061] When board C is produced in component mounting system **1**, an error may occur in any of component mounters **6**. For example, when an error exceeding a predetermined threshold value (for example, a mispicking of electronic components **4** by nozzle **7**) occurs in component mounter **6b** under the situation illustrated in FIG. 5, arithmetic device **60** displays image P6 illustrating the content of the process to be executed to resolve the error preferentially over images P2 to P4 as illustrated in FIG. 10. In the present example, image P6 is displayed above image P2 and image P3, and the display of image P4 is stopped. Image P6 includes a character string “Error Has Occurred” indicating that an error has occurred, a character string “Nozzle Exchange Operation” indicating a process to be executed to resolve the error, and an operation location “Component Mounter 6b”. In such a configuration, the user can easily know that an error has occurred, and can quickly address the error.

Correspondence Relationship

[0062] Interface device **52** and arithmetic device **60** are examples of a “display section” and a “control section”, respectively. Management device **8** is an example of a “display device”. The feeder carry, the feeder replacement, and the feeder return illustrated in FIG. 5 are examples of a “first process”, a “second process”, and a “third process”, respectively. Image P1, image P2, and image P3 illustrated in FIG. 5 are examples of a “first image”, a “second image”, and a “third image”, respectively. Image P4 illustrated in FIG. 7 is an example of a “first selection image”. Detail display button B3 is an example of a “second selection image”. Image P6 in FIG. 10 is an example of a “fourth image”.

[0063] Although the specific example of the technique disclosed in the present description is

described in detail, the specific example is merely an example, and does not limit the claims. The technique described in the scope of claims includes various modifications and changes of the specific examples described above. Hereinafter, modified examples of the above-described embodiment will be listed.

[0064] In the above-described example, an example in which the content of the operation for component mounter **6** in component mounting line **2** is displayed has been described. However, for example, the operation content for solder printer **5** incorporated in component mounting line **2** may be further displayed. Specifically, for example, processes such as a replenishment operation of a cream solder or an exchange operation of a mask may be further displayed.

[0065] In FIGS. **5** to **7**, arithmetic device **60** gradually increases the content of the nozzle exchange operation as the start time of the nozzle exchange operation approaches, displaying the content as image **P4**. Similarly, as the start times of the processes illustrated in image **P2** and image **P3** approach, arithmetic device **60** may gradually increase and display the contents of these processes as images **P2** and **P3**. Further, in place of the format gradually increasing the content of the nozzle exchange operation as the operation start time of the nozzle exchange operation approaches, for example, arithmetic device **60** may display image **P4** such that the luminance or the brightness of image **P4** gradually increases.

[0066] In addition, in the example described above, arithmetic device **60** displays the content of the operation to be executed by the user on interface device **52** of management device **8**, but the content may be displayed on interface device **24** of each component mounter **6** or on interface devices (not illustrated) of other working machines incorporated in component mounting line **2**, or may be displayed on a device separate from these devices.

[0067] Technical elements described in the present description or the drawings exhibit technical usefulness alone or in various combinations, and are not limited to combinations described in claims as filed. In addition, the technique exemplified in the present description or the drawings simultaneously achieves multiple purposes, and achieving one of the purposes itself has technical usefulness.

Claims

1. A display device comprising: a display section; and a control section, the display device being for displaying an operation to be executed by a user in a component mounting line on a process-by-process basis on the display section, wherein the control section is configured to, when a first image illustrating a content of a first process to be currently executed by the user is being displayed on the display section and when there exists a second process to be executed by the user after the first process, display a second image for guiding the second process on the display section along with the first image.
2. The display device according to claim 1, wherein the second image illustrates only a part of a content of the second process.
3. The display device according to claim 2, wherein the part of the content of the second process is one or multiple elements selected from a process name, an operation name, an operation location, an operation start time, and an operation content of the second process.
4. The display device according to claim 2, wherein the control section is configured to, as the operation start time of the second process approaches, change and display a content of the second image in a format in which the user more easily recognizes the content of the second image.
5. The display device according to claim 4, wherein the control section is configured to, as the operation start time of the second process approaches, increase the content of the second process to be displayed as the second image.
6. The display device according to claim 1, wherein the control section is configured to, when a third process which is different from the second process and which is to be executed by the user

exists after the first process and either simultaneously with or after the second process, display a third image for guiding the third process on the display section along with the first image and the second image

7. The display device according to claim 6, wherein the control section is configured to, when the operation start time of the second process and an operation start time of the third process are the same, assign priority, which is to be executed preferentially, to the second process and the third process, and display an image for guiding a process assigned a first priority in a format in which the user more easily recognizes the image for guiding the process assigned the first priority than an image for guiding a process assigned a second priority.

8. The display device according to claim 1, wherein the second image includes a first selection image for the user to select to execute the second process before the first process, and the control section is configured to, when the first selection image is selected by the user, display the second image in a format in which the user more easily recognizes the second image than the first image.

9. The display device according to claim 1, wherein the control section is configured to, when the first image and the second image are being displayed on the display section, further display a second selection image for the user to select to display a detailed image illustrating a list of operations to be executed by the user in the future on the display section, and when the second selection image is selected by the user, display the detailed image on the display section.

10. The display device according to claim 1, wherein the control section is configured to, when an error occurs in the component mounting line while the first image is being displayed on the display section, display a fourth image illustrating a content of a process to be executed by the user to resolve the error on the display section in a format in which the user more easily recognizes the fourth image than the second image.
